

TRAFFIC LIGHT DEMO PROGRAM IN 'C'



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Stoplights, also known as stop-and-go lights, traffic lights or traffic lamps, are signaling devices installed at road intersections that give a visual indication to control the flow of traffic. These signal the drivers on road when to stop, when to slow down and when to proceed, by utilising the combination of red, yellow and green lights. The lights are signaled in a particular sequence.

The red light is placed above the green light with yellow in between the two. The red signal indicates the vehicle to stop, yellow indicates to get ready and green indicates the vehicle to move forward. In some places, a digital count-down timer is employed along with the lights.

The above concept has been taken into consideration while developing this program to demonstrate the working of traffic lights. The project demonstrates working of the digital count-down timer, tri-colour stoplights and message display for the driver.

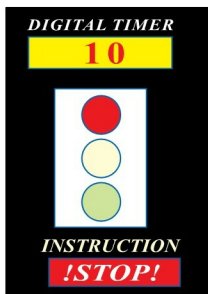
The digital count-down timer is set for ten seconds here and counts from 10 seconds down to 0 for the red light to turn on.

Tri-colour stoplights glow and go off one by one as the universally adopted traffic lights display (red, yellow and green lights, sequentially).

The instructions for the road users, such as stop, ready and go corresponding to glowing of the red, yellow and green lights, are simultaneously displayed in the instruction window at the bottom of the traffic lamps.

Software program

The software program is written in 'C' language and compiled using Turbo C IDE. Turbo C has a good collection



of graphics libraries. This program uses various 'C' graphics files such as `egavga.bgi`, `scri.chr`, `trip.chr` and `tscri.chr`. The implementation of graphics functions in this program can be used in any application with a few changes in the code.

Some of the key functions used are outlined below.

Initgraph(&gd, &gm, "c:\turboc3\bgi"). *Initgraph* initialises the graphics system by loading a graphics driver from the disk and then putting the system into graphics mode. It also resets all graphics settings like colour, palette and current position to their defaults.

'gd' specifies the graphics driver to be used.

'gm' specifies the initial graphics mode. When the program detects a graphics driver using 'gd = DETECT' function, the *initgraph* sets the graphics mode to the highest resolution available for the detected driver.

Closegraph(). Switches the screen from graphics mode back to text mode. This function is called after *getch()* function so that the screen doesn't clear

until the user hits the key.

Setbkcolor(number). Sets the background colour of choice from 16 colour options of 0 through 15. Instead of specifying the number, the respective colour name can also be used in block letters from the specified set.

SetColor(number). Selects a foreground or drawing colour from the same set of 16 colours.

SetTextstyle(font, direction, character size). Sets the values for the text characteristics to be used for displaying the text.

Outtextxy(x, y, "Literal expression"). Displays the string in graphics mode. (x, y) denotes the starting coordinate of the string on the screen.

Floodfill(). Fills the required colour in geometrical figures. Geometrical figures can be drawn using the inbuilt functions of shapes (rectangle, circle, etc).

Setfillstyle(pattern, color). This function too is used to fill the colour. It provides an additional feature of filling the colours in a specific pattern.

When the program is run, initially, the red light glows for ten seconds. Simultaneously, the instruction window shows 'Stop' message for the driver to stop and the 10-second countdown timer is displayed. When the counter reaches zero, the red light goes off and immediately the yellow light glows for a few seconds along with 'Ready' message displayed at the bottom. At this time, the driver can start the engine to get ready to move ahead. As soon as the yellow light goes off, the green light glows and simultaneously the 'Go' message is displayed in the instruction window at the bottom of the traffic lights.

EFY note. The source code and relevant graphics files are included in this month's EFY-CD and are also available on www.efymag.com website.